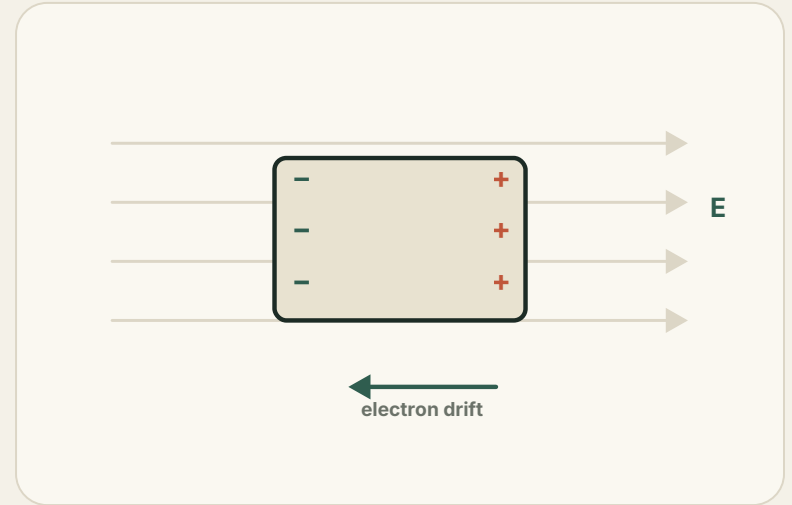


Static Electricity: Control & Use

How moving charges reach balance, create hazards, and do useful work.

ESSENTIAL QUESTION

How can the same electric charge be dangerous, safely blocked, or deliberately useful?



● LESSON MAP

Follow charge from motion to purpose.

01

BALANCE

Free electrons move until the field inside a conductor is zero.

02

CONCENTRATE

Charge crowds at sharp points and can ionise air.

03

SHIELD

A conducting shell redirects charge around a protected region.

04

APPLY

Electric forces route dust, paint, and toner.

THE IDEA

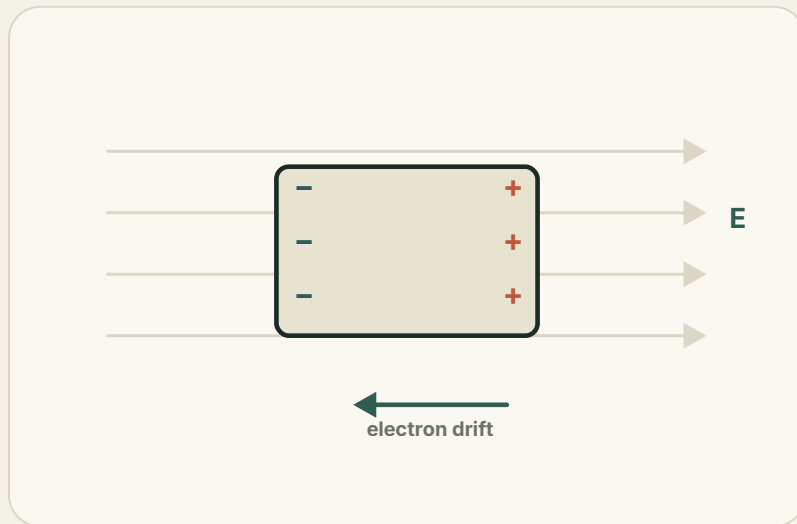
Four ideas explain both electrostatic **safety** and electrostatic **technology**.

OBSERVE

Free electrons move when a conductor enters a field.

The applied field pushes mobile electrons opposite to the field direction.

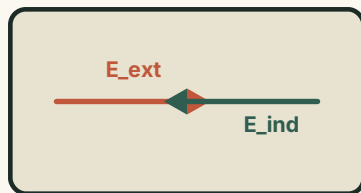
- 1 The external field acts**
Free electrons begin to drift through the metal.
- 2 Opposite charges appear**
At opposite ends — this is electrostatic induction.
- 3 They build an opposing field**
Which slows further movement.



MODEL

Charge stops moving when the internal field is zero.

Electrostatic equilibrium is a balance of fields, not a disappearance of charge.



net field zero — drift stops

INSIDE THE CONDUCTOR

$$E_{net} = E_{ext} + E_{ind} = 0$$

With no net field, electrons have no reason to keep drifting — equilibrium is reached extremely quickly.

CHECK

Equilibrium leaves four clear signatures.

These rules let us predict what every conductor will do.

01 · No field inside

$E = 0$ everywhere in the conducting material.

02 · One potential

The whole conductor is an equipotential.

03 · Surface charge

Any excess charge sits on the outer surface.

04 · Normal field

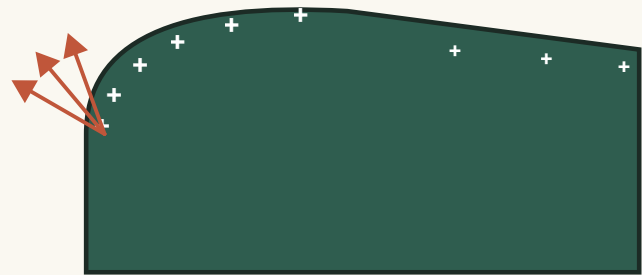
Just outside, the field is perpendicular to the surface.

COMPARE

Charge crowds where a surface is sharp.

A small radius of curvature means a larger surface charge density and a stronger nearby field.

- 1 Sharper point**
Smaller radius of curvature R .
- 2 More surface charge**
Higher charge per unit area σ .
- 3 Stronger nearby field**
Strong enough to affect the air.



Sharp point $\rightarrow$ crowded charge $\rightarrow$ strong field

APPLY

A sharp point can turn air into a conductor.

A strong field ionises nearby air; moving ions then carry charge away.

01 · Use it

Lightning rods give a controlled, grounded path.

02 · Prevent it

High-voltage conductors are smooth to reduce leakage.

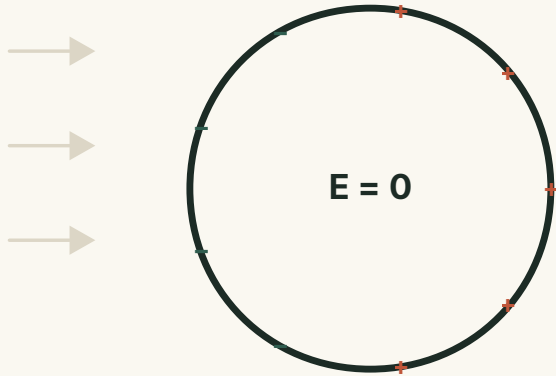
03 · Recognise it

Gas igniters and corona glow use the same effect.

SHIELD

A conducting shell keeps an external field out.

Surface charges rearrange until their field cancels the external field inside the cavity.



FARADAY-CAGE LOGIC

The shell does not block field lines like a wall — its mobile charges create the **cancelling** field.

Inside the cavity, $E = 0$.

COMPARE

Grounding decides whether the field leaks out.

A conductor shields its material; grounding can also remove charge from its outer surface.

1

Isolated shell

Inner charge induces equal opposite charge inside and equal charge outside.

1

Grounded shell

Ground supplies or removes charge, so the outside field can vanish.

SAFETY

Field-free interiors protect people and signals.

The same equilibrium principle scales from a cable to a live-line worker.

01 · Conductive suit

Metal fibres form a shielding layer around the worker.

02 · Vehicle body

A closed metal body routes lightning current around passengers.

03 · Cable screen

A grounded braid keeps electric noise off the signal.

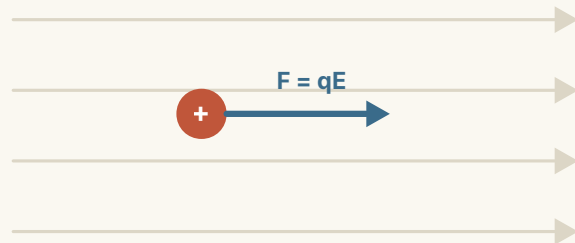
An electric field can steer tiny charged particles.

Once a particle carries charge, its path can be designed with the electric force.

GOVERNING RELATION

$$F = qE$$

More charge, or a stronger field, gives a larger steering force.

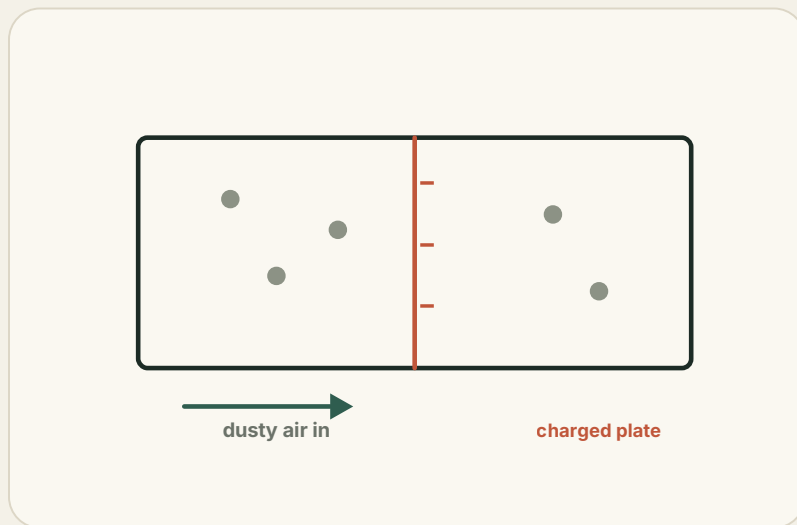


APPLY

A precipitator pulls pollution from the air.

Charge the dust, create a field, then collect particles on oppositely charged plates.

- 1 Charge the dust**
Ionised air transfers charge to particles.
- 2 Drive it sideways**
Electric force moves dust out of the airflow.
- 3 Remove and repeat**
Clean the collector plates periodically.



Electrostatic coating makes particles find the target.

Charging paint or toner improves coverage and reduces waste.

1

Spray painting

Charged droplets wrap around an oppositely charged workpiece.

1

Photocopying

Light writes a charge pattern; toner follows it before transfer to paper.

● SUMMARY

Control where charge gathers, where fields exist, and which path charges follow.

01 · Prevent

Use smooth surfaces and grounded paths to avoid damaging discharge.

02 · Protect

Use conducting shells and meshes to create field-free interiors.

03 · Use

Charge particles and steer them with $F = qE$.